

**Time:** 5 minutes**Outcome:** Calculate and interpret real and nominal interest rates.**Difficulty:** ●●○ Intermediate

Real versus Nominal Interest Rates



THE CORE IDEA

The real interest rate approximately equals the nominal interest rate minus inflation. The nominal rate states how many additional dollars a lender receives, while the real rate measures the change in purchasing power. Expected inflation determines the ex ante real rate when a loan is made; actual inflation determines the ex post real return and can redistribute purchasing power between borrowers and lenders.



HOW TO RECOGNIZE IT

- Unexpected inflation is higher than lenders and borrowers expected on a fixed-rate loan.
- Actual inflation is lower than expected on a fixed nominal loan.
- Unexpected inflation rises on a fixed-rate loan.
- Actual inflation exceeds expected inflation while a loan's nominal rate is fixed.
- Ex ante real interest uses expected inflation; ex post real interest uses actual inflation.



WATCH OUT

Do not use actual inflation to describe an expected return or expected inflation to describe the realized return. Ex ante real interest uses expected inflation; ex post real interest uses actual inflation.



WORKED EXAMPLE: CALCULATING THE REAL INTEREST RATE

Nominal interest is 9% and inflation is 5%. Using the usual approximation, real interest is $9\% - 5\% = 4\%$. The exact real return is $(1.09 / 1.05 - 1) \times 100 = 3.81\%$. Higher-than-expected inflation lowers the realized real repayment burden and shifts purchasing power from lender to borrower.



CHECK YOURSELF

What does a real interest rate describe that a nominal rate does not?

**READY?**

Return to the game and master this concept.